

Which of the following element groups is NOT included in the representative elements?

- ☐ A. Group 1 [2%]
- ✓ ☒ B. Group 3 [61%]
- ☐ C. Group 13 [12%]
- ☐ D. Group 18 [23%]

Correct

61% answered correctly

Explanation:

The diagram shows a periodic table with the following labels and groupings:

- Representative elements:** Indicated by a bracket at the top, it includes Groups 1, 2, and 13 through 18.
- Groups:** Labeled at the top of the table: 1 (IA), 2 (IIA), 3 (IIIB), 4 (IVB), 5 (VB), 6 (VIB), 7 (VIIB), 8 (VIII), 9 (VIII), 10 (VIII), 11 (IB), 12 (IIB), 13 (IIIA), 14 (IVA), 15 (VA), 16 (VIA), 17 (VIIA), and 18 (VIIIA).
- Blocks:** Labeled at the bottom: s-block (Groups 1-2), d-block (Groups 3-10), and p-block (Groups 13-18).
- Element Categories:** Labeled within the table: Alkali metals (Group 1), Alkaline earth metals (Group 2), Transition metals (Groups 3-10), Pnictogens (nitrogen group) (Group 15), Chalcogens (oxygen group) (Group 16), Halogens (Group 17), and Noble gases (Group 18).

The **representative elements** (also called the **main group elements**) are the elements in **Groups 1–2** and **Groups 13–18** on the periodic table. These elements are so called because they are the most abundant elements in nature and provide a wide representation of the elemental and chemical characteristics found in natural systems.

With respect to atomic **electronic configurations**, the representative elements are those elements found in the **s-block** and the **p-block** of the periodic table. The s-block elements have valence

electrons of highest energy occupying an s orbital whereas the p -block elements have valence electrons of highest energy occupying a p orbital. An **alternative numbering system** for the element groups involves a number (often expressed as a Roman numeral) along with a letter (A or B). Although this older numbering system is now considered obsolete, it is sometimes still used because, for the representative elements, it correlates to the number of **valence electrons in each group** (with the exception of helium). For example, elements in Group 3A each have three valence electrons.

(Choice A) Group 1 (1A) is part of the s-block elements and is included in the representative elements.

(Choices C and D) Group 13 (3A) and Group 18 (8A) are part of the p -block elements and are included in the representative elements.

Educational objective:

The representative elements (main group elements) are the elements in Groups 1–2 (1A–2A) and Groups 13–18 (3A–8A) that are contained in the s-block and the p -block, respectively, of the periodic table.

Time spent:0 sec

Subject:
General Chemistry

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System:
Atomic Theory and Chemical Composition